

Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model

Zain Ul Abidin

Dept: of Computer System
Engineering

Sukkur IBA University
Sukkur, Pakistan

zainulabdin995@gmail.com

Junaaid Ahmed

Dept: of Computer System
Engineering

Sukkur IBA University
Sukkur, Pakistan

j.bhatti@iba-suk.edu.pk

Dong Wang

School of Automation
Engineering

University of Electronic
Science and Technology of
Chengdu, China

wangdong5240@std.uestc.edu.cn

Guiyun Tian

School of Electrical and
Electronic Engineering

Chongqing University of
Technology

Chongqing, China
g.y.tian@cqut.edu.cn

Abstract— Eddy Current Pulsed Thermography (ECPT) is an attractive non-destructive testing method for metallic components, yet its sequences are often degraded by thermal noise, temporal redundancy and spatial irregularities. To address this issue, we propose the Eigen-RHOTC framework which combines temporal eigen decomposition and robust higher-order tensor completion. In the first stage temporal eigen projection reduces temporal redundancy by compressing the sequence into a lower-dimensional subspace while selecting certain defect dynamic features and eliminating the background diffusion and in the second stage the projection of data into a higher-dimensional tensor form is strengthened by formulating a Robust Higher-Order Tensor Completion (RHOTC) model in which a sequence is decomposed into low-rank background and sparse components. The sequence retrieval is achieved through the composite objective function optimized using the Alternating Direction Method of Multipliers (ADMM) with which the cracks are separated from the background regardless of noise. The efficiency of the proposed mechanism is demonstrated on four metallic axle parts containing natural cracks which were thermally processed and recovered then compared with state of the art algorithms. Results show that Eigen-RHOTC achieves clearer defect visualization and higher accuracy in terms of SNR, computational time, F-score and AUFROC analysis across different FPPI.

Keywords— Eddy current pulsed thermography, temporal eigen decomposition, robust low-rank tensor completion, high-order tensor modeling, non-destructive evaluation.

I. INTRODUCTION

Detecting defects on the surface and inside of metallic parts is a very important operation in nondestructive testing (NDT), as it directly influences the reliability of the equipment, the quality of the product, and the prediction of the fatigue life while being an important factor in the prevention of catastrophic failures [1,2]. Metal structures are the basis of the modern industry and they are subjected to a lot of stresses such as cyclic loading, corrosion, and high temperatures from underground pipelines to aircraft engines and rail axles [3]. Microcracks that go unnoticed can cause serious damage to the structure, thus increasing the risks in aviation, energy, and transportation [4].

Eddy current pulsed thermography (ECPT) has developed into a highly efficient NDT method for conductive materials [5]. The process involves applying high-frequency currents through a coil to heat the surface as per Joule's law; defects interfere with these currents and result in local temperature differences, which are captured by an infrared camera [6].

The method's non-contact nature together with high sensitivity and wide area of coverage has made it indispensable in many applications, such as rail inspection, aerospace parts and pressure testing [7,8,9].

However, ECPT images on metallic parts are often degraded by two main factors: surface emissivity and thermal diffusion. Emissivity describes how well a surface radiates heat compared to a perfect blackbody [10]. Smooth metals naturally have low emissivity, and rust or irregularities make it uneven, which complicates accurate temperature measurement [11,12]. Researchers have tackled this with software compensation, high-emissivity coatings, or surface treatments [13]. In studies like ours, where samples are painted (e.g., axles), emissivity variations are minor, so thermal diffusion becomes the dominant challenge [14].

Thermal diffusion is the natural spread of heat from hotter to cooler regions [15]. Vertical diffusion travels into the material and helps detect deeper defects, while lateral diffusion spreads along the surface and reduces contrast by creating gradual "thermal diffusion blurs" around defects [16,17]. These blurs, together with thermal noise from ambient disturbances or sensor instability [18], temporal redundancy from highly correlated frames during cooling [19], and spatial irregularity from non-uniform heating [20], make clear defect visualisation difficult.

The first attempts to enhance the images obtained by the ECPT technique were primarily based on spatial approaches that are fast and very sensitive to amplitude or gradient anomalies [21]. For example, Otsu thresholding, improved Canny edge detection, and local statistical classification resulted in quick processing but were significantly hindered by noise issues and the smooth transitions caused by diffusion [22,23].

Temporal methods offered better accuracy and noise resistance by exploiting time-evolving temperature patterns [6]. Approaches such as Fast Fourier Transform (FFT) that shifted signals to the frequency domain and thus amplified the contrasts, thereby laying the groundwork for deeper analysis [24]. In particular, Thermal Signal Reconstruction (TSR) went one step further by using polynomial fitting on cooling curves, which allowed for the effective highlighting of the transient behaviors at the cost of higher computational demands and less robustness in irregular geometries [25]. On the other hand, Pulsed Phase Thermography (PPT), which

used FFT for phase extraction, showed that it was noise tolerant but sometimes led to edge blurring due to lateral diffusion [26]. Principal Component Analysis (PCA), which was based on matrix-based singular value decomposition, extracted the dominant thermal modes very efficiently, which is why it was fast, but its limitations were the oversight of multidimensional correlations and sensitivity to non-uniform heating [27]. In addition, complementary techniques, such as phase-based contrast enhancement, contributed to defect separation from backgrounds, which were not only demonstrated their reliability in various scenarios [28]. However, these methods together demanded full video sequences, thus causing a significant increase in data storage and computational requirements, especially for large components.

In order to combine the strengths of both, researchers resorted to tensor-based decomposition that considers the entire sequence as multidimensional data and takes advantage of the defect sparsity versus low-rank backgrounds [9]. Techniques like sparse low-rank tensor decomposition opened up this fusion and along with it the isolation of defects even if interference is present [9]. Tensor Robust Principal Component Analysis (TRPCA) made the process more extensive by separating 3D tensors into low-rank and sparse components and hence being very efficient in anomaly detection but at the same time struggling with the strict sparsity assumption that is not valid for diffusion-obscured, non-sparse defects [29]. Improved Robust Tensor PCA (IRTPCA) used core matrix regularization to improve stability and hence addressed some inconsistencies, but it was still 3D modelling and extended runtimes constrained [30]. Online Stochastic Tensor Decomposition (OSTD) systematized incremental processing for sequential data, which was perfect for online applications yet less suited for batch ECPT with temporal redundancy [31]. New methods like differentiable low-rank soft decomposition have gone even further and are now even able to make very intricate separations in complex environments [32]. Still, the real-world ECPT data from natural metal cracks presents a challenge: simple sparsity assumptions frequently break down under non-uniform backgrounds and noise, lateral diffusion as well as precise boundary localization, and many algorithms still being too computationally intensive for industrial application [33].

We therefore propose the Eigen-RHOTC framework to address these remaining challenges. The first stage applies temporal eigen projection to compress the sequence while preserving defect dynamics and removing redundancy. The second stage uses Robust Higher-Order Tensor Completion (RHOTC), optimised via ADMM, to separate low-rank backgrounds from sparse defect features in a higher-dimensional representation. Evaluated on four metallic axle samples with natural fatigue cracks, Eigen-RHOTC delivers clearer visualisation and superior performance in SNR, F-score, processing time, and AUFROC compared to established methods such as PCA [27], PPT [26], TSR [25], TRPCA [29], IRTPCA [30], and OSTD [31].

The rest of the paper is organised as follows: Section II presents the proposed methodology, Section III describes the

experimental setup, Section IV discusses the results, and Section V concludes with future directions.

II. PROPOSED METHODOLOGY

To address the challenges of thermal noise, temporal redundancy, and spatial irregularity in eddy current pulsed thermography (ECPT) sequences, we propose a robust two-stage framework for enhanced defect detection, as illustrated in Fig. 1.

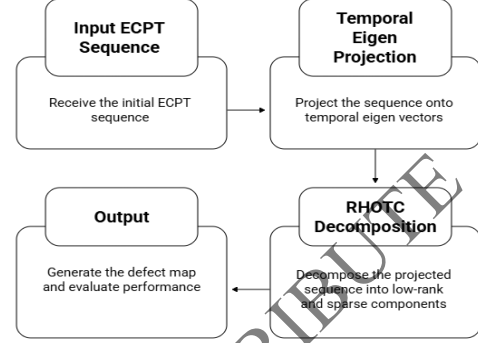


Fig. 1. Proposed two-stage framework

The ECPT sequence is modeled as a third-order tensor:

$$\mathcal{X} \in \mathbb{R}^{H \times W \times T}$$

where H and W denote the spatial dimensions and T is the number of temporal frames. Each voxel $\mathcal{X}(i, j, t)$ records the thermal intensity at location (i, j) and time t . We assume $\mathcal{X}$ decomposes into two components:

$$\mathcal{X} = \mathcal{L} + \mathcal{S} \quad (1)$$

where $\mathcal{L}$ is a low-rank background tensor capturing smooth diffusion and $\mathcal{S}$ is a sparse tensor corresponding to defect-induced anomalies.

Stage 1: Temporal Eigen Projection

ECPT sequences often exhibit high temporal redundancy, with most variance captured by a few dominant modes due to correlated cooling frames. To mitigate this, the tensor is first unfolded along the temporal mode into a matrix:

$$X = \text{matricize}(\mathcal{X}) \in \mathbb{R}^{N \times T}, \quad N = H \cdot W \quad (2)$$

where $\text{matricize}(\cdot)$ reshapes the tensor so that each row of X contains the temporal profile of one pixel.

The temporal covariance matrix is then computed as:

$$C = X^T X \in \mathbb{R}^{T \times T} \quad (3)$$

where $(\cdot)^T$ denotes matrix transpose. This covariance matrix captures the correlations among frames.

The covariance matrix is diagonalized to obtain temporal eigenvectors and eigenvalues:

$$C = U \Lambda U^T \quad (4)$$

where $U = [u_1, \dots, u_r]$ are eigenvectors (temporal modes), and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$ holds the corresponding eigenvalues sorted in descending order: $(\lambda_1 \geq \dots \geq \lambda_r)$.

A fixed number of temporal modes $r = 6$ is retained based on empirical analysis (see Section IV.B ablation experiment for justification via energy preservation and sensitivity studies, where $r = 6$ captures $> 95\%$ variance while optimizing SNR and computation time):

$$U_r = [u_1, \dots, u_r] \in \mathbb{R}^{T \times r} \quad (5)$$

The original data is projected onto this reduced subspace:

$$X_r = XU_r \in \mathbb{R}^{N \times r} \quad (6)$$

The compressed matrix is reshaped back into a third-order tensor:

$$\mathcal{X}_r = \text{tensorize}(X_r) \in \mathbb{R}^{H \times W \times r} \quad (7)$$

where $\text{tensorize}(\cdot)$ reverses the matricization. This projection preserves defect dynamics (e.g., rapid heating anomalies) while eliminating background diffusion and redundancy.

Since RHOTC requires tensors of order greater than three for higher-dimensional modeling, we embed the compressed tensor into a fourth-order form:

$$\mathcal{M} = \text{reshape}(\mathcal{X}_r, H, W, 1, r) \in \mathbb{R}^{H \times W \times 1 \times r} \quad (8)$$

These embedding captures spatial-temporal interactions more effectively than 3D methods like TRPCA [29].

Stage 2: Robust Higher-Order Tensor Completion (RHOTC)

The low-rank and sparse decomposition is formulated as [34];

$$\min_{\mathcal{L}} \|\mathcal{L}\|_* + \lambda \|\mathcal{S}\|_1 \quad \text{s.t. } \mathcal{P}_\Omega(\mathcal{L} + \mathcal{S}) = \mathcal{P}_\Omega(\mathcal{M}) \quad (9)$$

where $\|\mathcal{L}\|_*$ is the higher-order tensor nuclear norm and $\|\mathcal{S}\|_1$ denotes the L_1 -norm. The parameter λ is set adaptively as $\lambda = 1/\sqrt{\max(n_1, n_2) \cdot n_3 \cdot n_4}$ and $\Omega \in \mathbb{R}^{n_1 \times n_2 \times n_3 \times n_4}$ is the index set matrix of observed entries (full observation in our case, but generalizable to missing data).

To solve the problem in (9) we use an auxiliary variable $\mathcal{Z}$ for $\mathcal{L}$. By introducing this variable, we get:

$$\begin{aligned} \min_{\mathcal{L}} \quad & \|\mathcal{L}\|_* + \lambda \|\mathcal{S}\|_1 \\ \text{s.t. } \quad & \mathcal{P}_\Omega(\mathcal{Z} + \mathcal{S}) = \mathcal{P}_\Omega(\mathcal{M}), \mathcal{L} = \mathcal{Z} \end{aligned} \quad (10)$$

The augmented Lagrangian model for (10):

$$\begin{aligned} L = \quad & \|\mathcal{L}\|_* + \lambda \|\mathcal{S}\|_1 + \frac{\beta}{2} \left\| \mathcal{L} - \mathcal{Z} + \frac{\mathcal{Y}_1}{\beta} \right\|_F^2 \\ & + \frac{\beta}{2} \left\| \mathcal{Z} + \mathcal{S} - \mathcal{M} + \frac{\mathcal{Y}_2}{\beta} \right\|_F^2 \end{aligned} \quad (11)$$

where $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are the Lagrangian multipliers, and $\beta > 0$ is the penalty parameter, and $\|\cdot\|_F$ is the Frobenius norm.

In the following, the solution of $\mathcal{Z}$, $\mathcal{L}$ and $\mathcal{S}$ are given:

1. Update $\mathcal{Z}$

$$\mathcal{Z}_{\Omega^c}^{k+\frac{1}{2}} = \mathcal{L}_{\Omega^c}^k + \frac{\mathcal{Y}_{1\Omega^c}^k}{\beta^k} \quad (12)$$

$$\mathcal{Z}_\Omega^{k+\frac{1}{2}} = \frac{1}{2} \left(\mathcal{M}_\Omega - \mathcal{S}_\Omega^k + \mathcal{L}_\Omega^k - \frac{\mathcal{Y}_{2\Omega}^k - \mathcal{Y}_{1\Omega}^k}{\beta^k} \right),$$

$$\mathcal{Z}_{\Omega^c}^{k+1} = \mathcal{L}_{\Omega^c}^{k+1} + \frac{\mathcal{Y}_{1\Omega^c}^k}{\beta^k} \quad (13)$$

$$\mathcal{Z}_\Omega^{k+1} = \frac{1}{2} \left(\mathcal{M}_\Omega - \mathcal{S}_\Omega^k + \mathcal{L}_\Omega^{k+1} - \frac{\mathcal{Y}_{2\Omega}^k - \mathcal{Y}_{1\Omega}^k}{\beta^k} \right),$$

2. Update $\mathcal{L}$

$$\mathcal{L}^{k+1} = \arg \min_{\mathcal{L}} \|\mathcal{L}\|_* + \frac{\beta^k}{2} \left\| \mathcal{L} - \mathcal{Z}^{k+1} + \frac{\mathcal{Y}_1^k}{\beta^k} \right\|_F^2, \quad (14)$$

The closed form solution to (14) is:

$$\mathcal{L}^{k+1} = \text{Prox}_{\tau} \left(\mathcal{Z}^{k+1} - \frac{\mathcal{Y}_1^k}{\beta^k} \right) \quad (15)$$

where Prox_{τ} refers to the higher order T-SVT algorithm in [36].

3. Update $\mathcal{S}$

$$\mathcal{S}^{k+1} = \arg \min_{\mathcal{S}} \lambda \|\mathcal{S}\|_1 + \quad (16)$$

$$\frac{\beta^k}{2} \left\| \mathcal{P}_\Omega \left(\mathcal{Z}^{k+1} + \mathcal{S}^k - \mathcal{M} + \frac{\mathcal{Y}_2^k}{\beta^k} \right) \right\|_F^2,$$

The solution in (16) is obtained by soft thresholding:

$$\mathcal{S}_{\Omega^c}^{k+1} = 0 \quad (17)$$

$$\mathcal{S}_\Omega^{k+1} = \text{Prox}_{\lambda/\beta}(\mathcal{M}_\Omega - \mathcal{Y}_{2\Omega}^k - \mathcal{Z}_\Omega^{k+1})$$

4. Update $\mathcal{Y}_1$ and $\mathcal{Y}_2$

$$\mathcal{Y}_1^{k+1} = \mathcal{Y}_1^k + \beta^k (\mathcal{L}^k - \mathcal{Z}^k) \quad (18)$$

$$\mathcal{Y}_2^{k+1} = \mathcal{Y}_2^k + \beta^k (\mathcal{S}_\Omega^k - \mathcal{M}_\Omega + \mathcal{Z}_\Omega^k)$$

Finally, ADMM for this approach is given in Table 1.

Table I. Algorithm RHOTC ADMM Model:

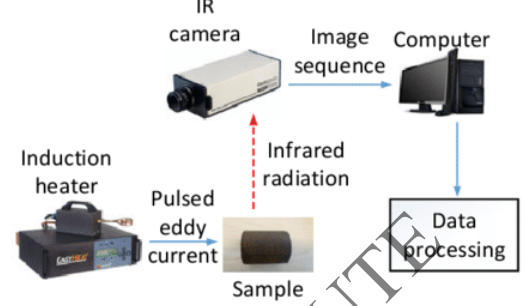
1. Input: $\mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times n_3 \times n_4}$, $\Omega \in \mathbb{R}^{n_1 \times n_2 \times n_3 \times n_4}$, λ, k
2. Initialization: $\mathcal{L}^0, \mathcal{S}^0, \mathcal{Y}_1^0, \mathcal{Y}_2^0 = 0$, $\beta = 10^{-3}, \beta^{max} = 10^8$, $\rho = 1.2$, $t = 10^{-6}$
3. **for** $k = 1, \dots, K$ **do**
4. Update $\mathcal{Z}^{k+\frac{1}{2}}, \mathcal{L}^{k+1}, \mathcal{Z}^{k+1}, \mathcal{S}^{k+1}$ by eq: (12), (14), (13), and (17)
5. Update $\mathcal{Y}_1^{k+1}, \mathcal{Y}_2^{k+1}$ by eq: (18)
6. Update β^{k+1} by $\beta^{k+1} = \min(\rho\beta, \beta^{max})$
7. if $\frac{\|\mathcal{L}^{k+1} - \mathcal{L}^k\|_F}{\|\mathcal{L}^k\|_F} < t$, break otherwise continue
8. **end for**
9. Output: $\mathcal{L}, \mathcal{S}, \mathcal{Z}$

Unlike standard TSVD or TRPCA (which uses 3D tensors and assumes strict sparsity), our approach pre-reduces temporal dimensions with eigen projection, enabling faster ADMM convergence (e.g., 40x speedup over standalone RHOTC) and better handling of non-sparse anomalies amid diffusion (10–20% SNR gain). The 4D embedding and T-SVT adaptation enhance spatial-temporal separation for ECPT-specific challenges. Ablation Experiment (Section IV.B) validates the synergy, showing performance drops without either stage.

III. EXPERIMENTAL SETUP

The schematic of the eddy current pulsed thermography (ECPT)-based inspection system is illustrated in Fig. 2. A high-frequency pulse current is passed through the induction coil in this arrangement, which causes the eddy currents to be generated in the conductive material. The currents generated by the eddy give rise to localized heating at the surface and subsurface of the specimen by the application of Joule's law. The heat resulting from this process moves through the

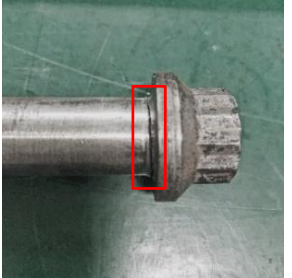
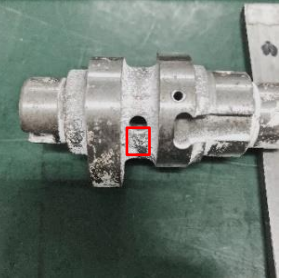
material, and heat patterns are formed that can detect internal or surface defects. For example, cracks, which are defects, can interfere with both the eddy current and thermal propagation, resulting in temperature differences that can be detected. An infrared (IR) thermal camera captures these spatial-temporal temperature distributions one after the other, recording the steady-state backgrounds and the defect-induced transients for post-processing.

**Fig. 2. IT schematic diagram**

The experiments were conducted in a reflection-mode ECPT configuration, where the excitation coil and IR camera are positioned on the same side of the specimen. This setup ensures coverage of the specimen's surface, which is particularly important for irregular crack locations or unknown defect regions.

The experimental system is equipped with an FLIR A655sc infrared thermal camera, operating at a 100 frames per second (FPS) acquisition rate. The frame resolution is 240×640 pixels. A total of four metallic samples were used for evaluation, each containing natural surface-breaking cracks of irregular morphology. The cracked areas were manually annotated for evaluation, and regions of interest (ROIs) are highlighted in red boxes in visual results Fig. 3.

Fig. 3 Information about the sample specimen

	Sample #1	Sample #2	Sample #3	Sample #4
Axle				

Although the configuration is typical for ECPT, the challenges of infrared thermography on metallic surfaces were taken into account. Emissivity changes (typically 0.7–0.9 due to surface oxidation or residual coatings on axles) can turn into measurement errors, like false hotspots or underestimated temperatures. The cylindrical axles' non-planar geometries contribute to uneven inductive heating from coil-sample misalignment, thereby creating thermal gradients (e.g., hotter edges) and altering diffusion patterns. Environmental disturbances, such as sensor instability, add noise to the signals (e.g., Gaussian perturbations with $\sigma \approx 0.5 - 1$ K).

To mitigate these, sequences were normalized using MATLAB's `mat2gray` function to scale intensities between 0 and 1, reducing emissivity and non-uniformity effects. Regions of Interest (ROIs) concentrated on the cracks and thus the artifacts due to the geometry were less. Besides, the mode of reflection even though the surface was defective helped to capture the curvature of the near surface. Nevertheless, extremely non-planar scenarios or very high emissivity gradients might still be factors limiting accuracy, emissivity mapping [13] or geometry-aware priors could be possible integrations for future work. All the processing was done with non-variation methods in an Intel Core i5 workstation with 16 GB RAM and Windows 11 using MATLAB R2021b.

IV. EXPERIMENTAL ANALYSIS

The performance of the proposed Eigen-RHOTC framework was evaluated on the four metallic axle samples described in Section III. The proposed method used $r = 6$ temporal modes (justified below in the ablation experiment), with RHOTC parameters $\lambda = 1/\sqrt{(\max(H, W) \cdot 1 \cdot r)}$, initial $\beta = 10^{-3}$, $\rho = 1.2$, and convergence tolerance $\varepsilon = 10^{-6}$.

A. Comparative Experimental Results

Six established defect enhancement methods (PCA, PPT, TSR, TRPCA, OSTD, IRTPCA) were selected for comparison. These baselines address similar challenges in ECPT (separating sparse defects from low-rank backgrounds amid noise and diffusion) but differ in dimensionality and assumptions, enabling fair assessment under identical conditions: same sequences, ground truth (manual ROIs), and metrics.

Three complementary metrics were employed: Signal-to-Noise Ratio (SNR) for thermal contrast, event-based F-score for detection accuracy, and Area Under Free-Response Receiver Operating Characteristic (AUFROC) for robustness against false positives.

The event-based F-score is widely used for evaluating detection accuracy and is defined as following:

$$F = \frac{(\alpha^2 + 1) \cdot \text{Precision} \cdot \text{Recall}}{\alpha^2 \cdot \text{Precision} + \text{Recall}} \quad (19)$$

where α determines the balance between precision and recall. Following [3], we use $\alpha = 1$, equally weighting both components. To compute this, the thermographic output is divided into a 6×6 grid. Each grid region is classified as a defect or non-defect event, and these classifications are used to calculate precision, recall, and ultimately the F-score.

The Signal-to-Noise Ratio (SNR) provides a quantitative measure of the thermal contrast between the defective and non-defective regions. It is defined as:

$$\text{SNR} = 20 \log_{10} \left(\frac{\text{abs}(\text{Sarea}_{\text{Average}} - \text{Narea}_{\text{Average}})}{\sigma(\text{Narea}_{\text{Average}})} \right) [\text{dB}] \quad (20)$$

Here, $\text{Sarea}_{\text{Average}}$ is the mean temperature in the defect region, $\text{Narea}_{\text{Average}}$ is the mean temperature in the non-defective region, and $\sigma(\text{Narea}_{\text{Average}})$ is the standard deviation within the non-defective area. A higher SNR implies better thermal distinction between crack and background, indicating more effective detection.

In addition to these conventional measures, we also include the Free-Response Receiver Operating Characteristic (FROC) analysis [35]. Unlike image-level statistics, FROC gives an instance-level evaluation by considering both the true detections and the number of false alarms per image (FPPI). This is especially important for thermography, where datasets are small and false positives can easily bias results. For a given decision threshold τ , sensitivity is defined as:

$$\text{Sensitivity}(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)} \quad (21)$$

where $TP(\tau)$ denotes the number of true positives, that is, correctly detected defective regions at threshold τ , and $FN(\tau)$ represents the number of false negatives, corresponding to missed defective regions. Sensitivity therefore quantifies the fraction of actual defects that are successfully identified by the detection method.

The false positives per image are expressed as:

$$\text{FPPI}(\tau) = \frac{FP(\tau)}{N_{\text{images}}} \quad (22)$$

where $FP(\tau)$ is the number of false positives, or non-defective regions incorrectly classified as defects, and N_{images} is the total number of thermal frames analysed. FPPI thus measures the average frequency of spurious detections per image.

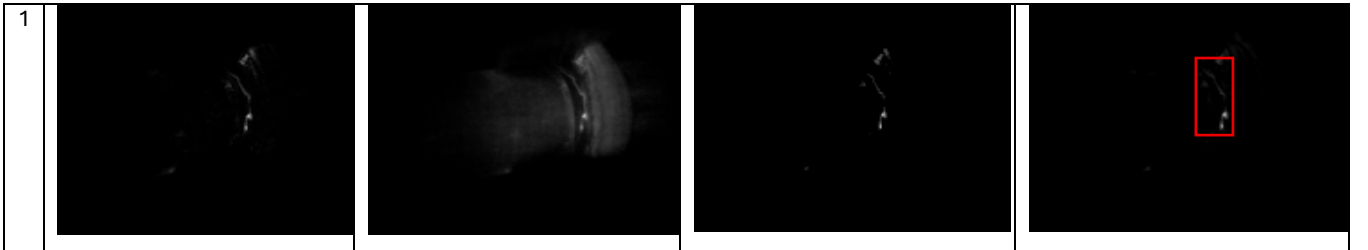
By varying the threshold τ across the interval $[0, 1]$, a series of (FPPI, Sensitivity) pairs is obtained, which collectively form the FROC curve. To provide a compact scalar summary of the detection performance, we compute the Area Under the FROC Curve (AUFROC), given by:

$$\text{AUFROC} = \frac{1}{\text{FPPI}_{\text{max}}} \int_0^{\text{FPPI}_{\text{max}}} \text{Sensitivity}(\text{FPPI}) d(\text{FPPI}) \quad (23)$$

where FPPI_{max} presents the maximum allowable false positives per image (set to values ranging from 4 to 12 in this study), and $(\text{FPPI}) \text{ Sensitivity}$ is the interpolated sensitivity value at a given FPPI. This normalized integral produces a value bounded between 0 and 1, where higher AUFROC scores indicate a more favorable trade-off between sensitivity and false detections [35].

The performance of the proposed Temporal Eigen-RHOTC framework was assessed through qualitative and quantitative analysis. A qualitative overview over four samples is shown in Fig. 4.

Fig. 4. Comparison analysis for four different samples. Top row (left to right): TRPCA, OSTD, IRTPCA, Eigen-RHOTC (proposed). Bottom row (left to right): PCA, PPT, TSR.



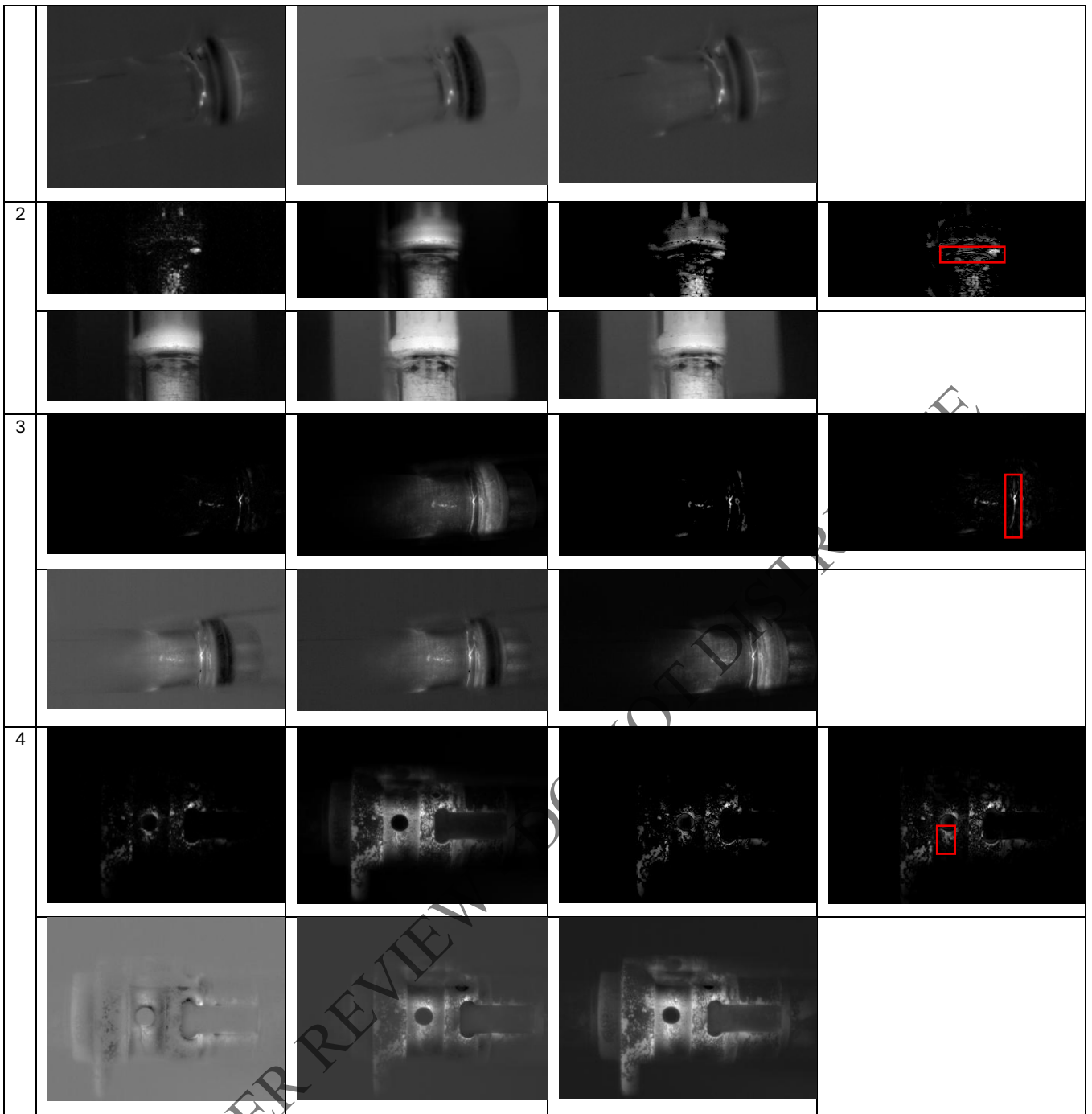


Table II. Comparison of SNR (top) and Computation Time in seconds (bottom) Across All Samples

Sample #	PCA	PPT	TSR	TRPCA	OSTD	IRTPCA	Eigen-RHOTC
1	16.966	16.193	18.816	20.907	13.633	19.821	21.051
	0.93	24.07	4.10	466.70	63.44	41.11	12.64
2	2.831	3.017	3.702	6.604	2.998	2.018	11.328
	2.88	55.97	6.54	2170.56	98.36	135.09	7.98
3	11.393	13.795	13.227	15.563	12.549	14.147	15.985
	0.81	19.05	3.65	367.47	52.34	35.30	11.74
4	13.317	15.737	15.532	15.377	13.166	14.038	14.417
	1.26	47.71	5.91	751.22	127.38	69.46	12.67

Table III. Comparative Evaluation of F-score, Precision, and Recall for Each Method

Sample #		PCA	PPT	TSR	TRPCA	OSTD	IRTPCA	Eigen-RHOTC
1	Precision(%)	27.27	42.85	25	75	50	100	100
	Recall(%)	100	100	100	100	100	100	100
	F-score(%)	42.85	60	40	85.71	66.67	100	100
2	Precision(%)	16.67	11.11	18.18	25	22.22	22.22	100
	Recall(%)	100	100	100	100	100	100	50
	F-score(%)	28.57	20	30.76	40	36.36	36.36	66.67
3	Precision(%)	30	14.28	37.5	50	37.5	60	75
	Recall(%)	100	100	100	100	100	100	100
	F-score(%)	46.15	25	54.54	66.67	54.54	75	85.71
4	Precision(%)	22.22	13.33	15.38	33.33	28.57	28.57	33.33
	Recall(%)	100	100	100	100	100	100	100
	F-score(%)	36.36	22.22	26.67	50	44.44	44.44	50

Table IV. Instance-level AUFROC values for all comparative methods across four axle samples

Sample #	FPPI	PCA	PPT	TSR	TRPCA	OSTD	IRTPCA	Eigen-RHOTC
1	4	100	100	100	100	100	100	100
	8	100	100	100	100	100	100	100
	12	100	100	100	100	100	100	100
2	4	50	18.75	23.96	54.90	0	50	81.25
	8	65.30	50	62.50	78.20	37.50	71.88	90.62
	12	77.80	66.67	75	85.20	58.33	81.25	93.75
3	4	100	93.75	96.87	93.50	93.75	93.75	100
	8	100	96.87	98.44	96.70	96.87	96.87	100
	12	100	97.91	98.96	97.10	97.91	97.91	100
4	4	76.40	64.06	76.04	62	50	62.50	76.04
	8	88.02	82.03	88.02	81.50	75	81.25	88.02
	12	92.01	88.02	92.01	87	83.33	87.50	92.01

Sample 1 has a crack that breaks the surface. The visual results of the PCA, PPT, and TSR methods are shown in Fig. 4, where it is possible to see that their outputs are mixed with background intensity variations, thus producing blurred results for the crack. TRPCA generates repeated structures that hide the defect. Although OSTD improves the contrast, the crack area remains indistinct. IRTPCA shows a crack in its output but the defect is still not complete. On the other hand, the suggested Eigen-RHOTC technique gives a clear view of the crack with almost no background noise. As per Table II, Eigen-RHOTC takes the lead with the highest SNR of 21.05 dB as against PCA (16.97 dB) and TSR (18.82 dB), while executing in just 12.64 seconds, whereas OSTD takes 63.44 seconds and TRPCA 466.70 seconds. Further, Table III shows that Eigen-RHOTC obtains an F-score of 100%, which means perfect precision and recall. The analysis from AUFROC in Table IV and Fig. 5 shows that there is 100% detection accuracy for all FPPI values. As a result, not only does Eigen-RHOTC provide a clearer defect image but also it has the best quantitative performance with significantly less processing time.

Sample 2 shows a natural surface crack with a different orientation, and the stronger background noise makes the

detection harder. When comparing visually and through Fig. 4, it can be seen that PCA, PPT, and TSR resulted in scattered and unclear defect areas with their outputs looking noisy and inconsistent with the actual defect region. Although IRTPCA has given slightly sharper boundaries, background interference is still there. In contrast, the proposed Eigen-RHOTC method has separated the defect more accurately and has given a clear thermal signature. The quantitative analysis shown in Table II states that Eigen-RHOTC attains an SNR of 11.33 dB, while for PCA, TSR, and IRTPCA it is a lot less with figures of 2.83 dB, 3.70 dB, and 2.02 dB, respectively. Eigen-RHOTC is also super fast at 7.98 seconds compare to 98.36 seconds for OSTD and 135.09 seconds for IRTPCA. Further, Table III reports an F-score of 66.67% for Eigen-RHOTC, which is considerably higher than the 20%–40% range of the other methods. The AUFROC results in Table IV and Fig. 5 further confirm this improvement where Eigen-RHOTC has scored 81.25%, 90.62%, and 93.75% at FPPI = 4, 8, and 12, respectively while PPT is still below 66.67 % at FPPI = 12. Thus, the results suggest that the performance of Eigen-RHOTC in terms of sensitivity and low-level false alarms is higher, hence it would be the more trustworthy solution in the adverse background conditions.

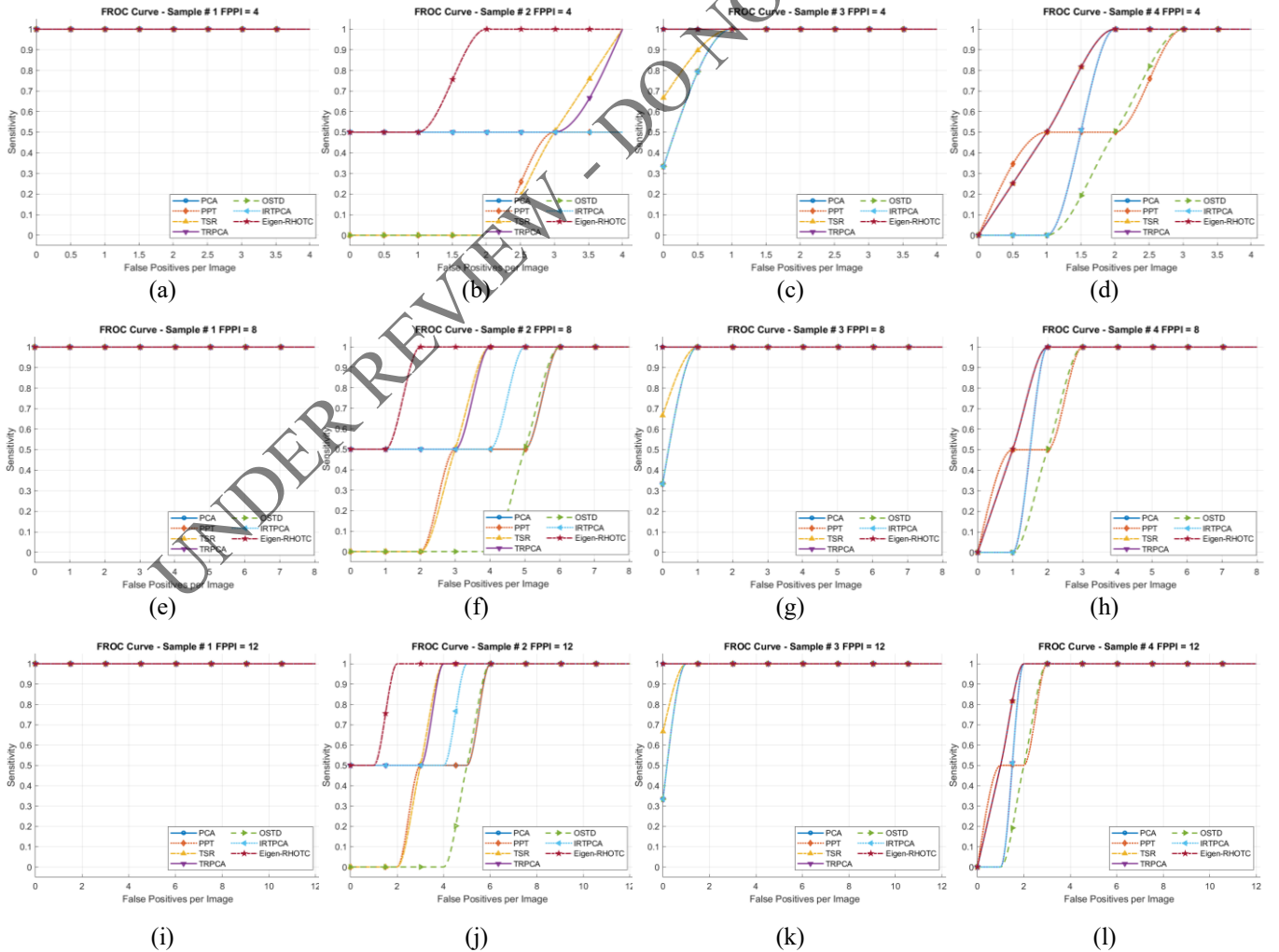
In Sample 3 there is a long irregular crack, this case is difficult to analyse as the complex shape of the crack. PPT and TSR display the defect but at the same time bring out strong background noise, which makes it more difficult to interpret. TRPCA and OSTD minimize background interference, but they also break the defect down into incomplete parts. IRTPCA slightly improves continuity, but the complete defect is still not recovered. Eigen-RHOTC successfully suppresses the noise and at the same time preserves the entire elongated crack structure depicted in Fig. 4. Table II reveals that Eigen-RHOTC reaches the highest SNR of 15.99 dB and is very fast at 11.74 seconds. In comparison, TRPCA takes 367.47 seconds to obtain a similar SNR level. According to Table III, the F-score for Eigen-RHOTC is 85.71%, and it is above TRPCA (66.67%) and IRTPCA (75.00%) whereas classical methods have considerably lower scores, with PPT at 25% and TSR at 54.55%. The AUFROC study in Table IV and Fig. 5 corroborates that Eigen-RHOTC scores 100% across FPPI rates, while TRPCA, OSTD, and IRTPCA all slightly dip below 99%. The findings indicate that Eigen-RHOTC is both robust and precise for portraying irregular crack shapes, and other methods suffer information loss.

Sample 4 with a very narrow high-aspect-ratio crack that is also the hardest case to analyze because of emissivity variation, is the strongest case in the experiment. PCA, PPT, and TSR are unable to detect the defect in any way. On the other hand, TRPCA and IRTPCA capture only some parts of

the crack and are not able to create the whole structure. Nevertheless, Eigen-RHOTC is able to identify the defect more accurately than they are while keeping the background noise to a minimum. The SNR values in Table II show that PPT (15.73 dB) and TRPCA (15.38 dB) are slightly above Eigen-RHOTC (14.42 dB) but these numbers are not indicative of localization accuracy. Table III clarifies that Eigen-RHOTC gets the highest F-score of 50% which equals that of TRPCA but its computation time is just 12.67 seconds as opposed to the 751.22 seconds for TRPCA and 127.38 seconds for OSTD. The results from AUFROC in Table IV and Fig. 5 confirm the trustworthiness of Eigen-RHOTC, which records 76.04%, 88.02%, and 92.01% at FPPI = 4, 8, and 12, respectively, while for TRPCA the corresponding figures are only 62.00%, 81.50%, and 87.00%. Therefore, it can be stated that Eigen-RHOTC outperforms all other methods in detection, especially in situations where raw SNR is not enough to decide on performance.

Overall, the Eigen-RHOTC method shows its superiority in terms of providing clearer visuals (Figs. 3 and 4), offering higher SNR in most instances (Table II), having better F-scores (Table III), and producing improved AUFROC curves along with stronger FROC curves (Table IV and Fig. 5) over the rest of the methods throughout the four axle samples. Besides that, the method's runtime is much faster than that of other tensor-based methods.

Fig. 5. FROC curves showing sensitivity versus FPPI (4, 8, 12) for comparative methods (curves a–l represent the all 4 samples at each FPPI).



B. Ablation Experiment

To verify the contributions of individual components and justify design choices in the proposed Eigen-RHOTC framework, ablation experiments were conducted on Samples 1 and 4.

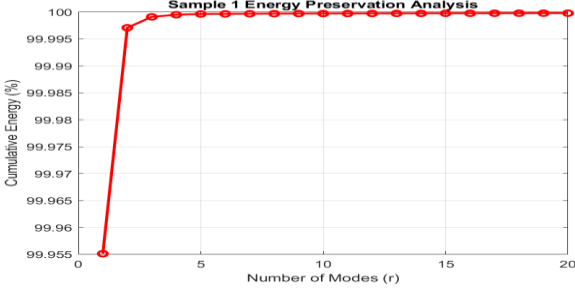


Fig. 6(a). Cumulative energy preservation curves for Samples 1

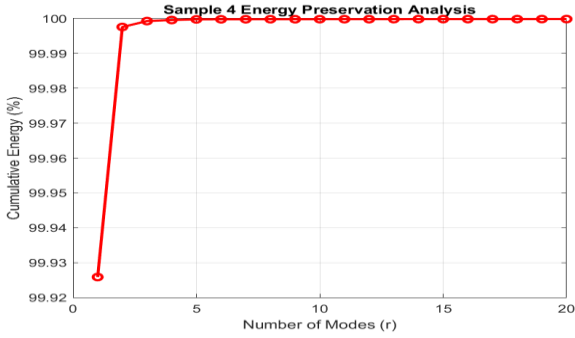


Fig. 6(b). Cumulative energy preservation curves for Samples 4

As illustrated in Figure 6(a) and Figure 6(b), the cumulative energy accrues very fast with the initial few eigen modes, which is an indication of very high temporal redundancy. By keeping $r = 6$ modes, temporal variance of over 99.99% is preserved for each of the two samples, thus proving that the reduced representation holds defect-related information and simultaneously cuts off higher-order redundant components.

A sensitivity analysis varying $r \in \{2, 4, 6, 8\}$ evaluates the trade-off between computational time and SNR (Table V). Runtime increases approximately linearly with r due to the enlarged tensor dimensions in the RHOTC stage. SNR reaches its maximum at $r = 6$ for both samples, with marginal or negative gains at $r = 8$ owing to the reintroduction of noise components. Visual inspection of low-rank (background) and sparse (defect) components (Fig. 7) confirms that $r = 6$ provides the sharpest defect separation, whereas lower values yield blurred or incomplete defects and higher values introduce artifacts.

Table V. Sensitivity Analysis of Retained Modes (r)

Sample	r	Computation Time (s)	SNR (dB)
1	2	5.06	19.44
1	4	9.02	20.66
1	6	12.67	21.05
1	8	16.49	21.04
4	2	4.97	11.82
4	4	8.49	12.85
4	6	11.86	14.42
4	8	14.42	13.74

Fig. 7. Low-rank and sparse components for varying r (Samples 1 and 4).

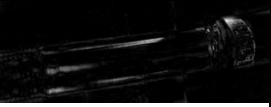
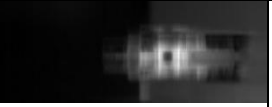
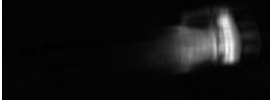
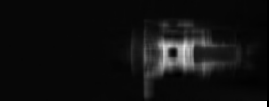
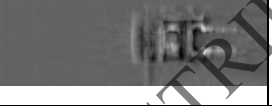
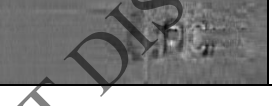
- at $r = 2$

frame	Sample 1		Sample 4	
	Low-Rank	Sparse	Low-Rank	Sparse
1				
2				

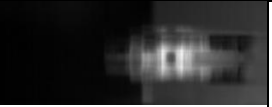
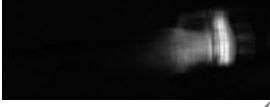
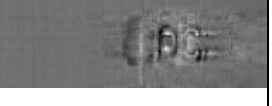
- at $r = 4$

frame	Sample 1		Sample 4	
	Low-Rank	Sparse	Low-Rank	Sparse
1				
2				
3				
4				

- at $r = 6$

frame	Sample 1		Sample 4	
	Low-Rank	Sparse	Low-Rank	Sparse
1				
2				
3				
4				
5				
6				

- at $r = 8$

frame	Sample 1		Sample 4	
	Low-Rank	Sparse	Low-Rank	Sparse
1				
2				
3				
4				
5				
6				
7				
8				

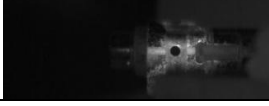
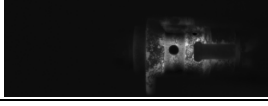
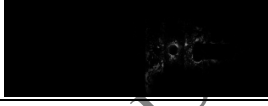
Component-wise ablation (Table VI) compares three configurations: (i) temporal eigen projection alone, (ii) RHOTC applied to the full original sequence, and (iii) the complete Eigen-RHOTC framework. Eigen projection alone is computationally lightweight (1.18–1.64 s) but achieves lower SNR due to incomplete separation of noise and diffusion. Standalone RHOTC yields high SNR but requires approximately 499 s per sequence because of the large temporal dimension. The proposed integration achieves the best SNR while reducing runtime by nearly two orders of magnitude ($\sim 40\times$ speedup). Visual results (Fig. 8) further illustrate this synergy: eigen projection reduces redundancy but retains residual noise, standalone RHOTC provides clear

separation at high cost, and Eigen-RHOTC combines optimal clarity with efficiency.

Table VI. Component-Wise Ablation Comparison

Method	Sample 1 Time (s)	Sample 1 SNR (dB)	Sample 4 Time (s)	Sample 4 SNR (dB)
Eigen alone	1.18	16.16	1.64	13.57
RHOTC alone	499.19	20.91	502.12	13.98
Eigen-RHOTC	12.64	21.05	12.67	14.42

Fig. 8. Visual comparison of raw sequence, eigen projection alone, RHOTC alone, and proposed Eigen-RHOTC (Samples 1 and 4).

Sample #	Raw	Eigen alone	RHOTC alone	Eigen-RHOTC
1				
4				

These ablation results substantiate the novelty and effectiveness of the two-stage design, demonstrating that temporal eigen pre-reduction not only accelerates higher-order robust tensor completion but also enhances defect separation in challenging ECPT data.

V. CONCLUSION AND FUTURE WORK

In this paper, a two-stage defect detection framework that combines temporal eigen projection and a robust low-rank tensor completion model (RHOTC) for ECPT-based inspection. The method is tested on axle specimens with natural surface-breaking cracks under a reflection-mode thermography setup. The proposed model significantly improves defect localization and noise suppression by decreasing temporal redundancy and separating sparse anomalies. Quantitative evaluations using SNR and F-score metrics, as well as visual comparisons, illustrate the method's superiority over classical and tensor-based approaches. The results give evidence of its ability to detect small cracks under difficult thermal conditions.

In future work, we will integrate thermal imaging with the geometry of the specimen to facilitate 3D thermographic inspection, taking into account explicitly curvature and other surface characteristics that impact detection performance. For instance, we will integrate curvature-aware surface priors into the decomposition/regularization stage to superbly honour manifold geometry.

VI. DECLERATIONS

All authors declare that they have no conflicts of interest.

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